CEO
M. Bilal Javaid

Valid from: 21th May 2026

Valid To: 28th May 2026

Feasibility Report

Development of Vantare VantageSuit™: A Real-Time Biomechanical Feedback Exosuit

Client: Mark

About M. Bilal Javaid

Link to Intro

- 8+ years of experience in industrial design and electronics engineering.
- Expertise in PCB design, 3D modeling, and Prototyping.
- Specialized in embedded systems (STM32, ESP32, Raspberry Pi)
- Proficient in CAD, Rendering and DFM
- Builds practical, manufacturable tech solutions by uniting design and engineering.

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Project Scope

The project focuses on the continued design and development of the Vantare VantageSuit™, an intelligent wearable exosuit platform intended for real-time biomechanical monitoring, movement analysis, posture correction, and adaptive performance feedback. The development effort is a continuation project and not a initiative; however, despite the availability of previous engineering assets, the current phase will begin with a fresh redevelopment approach to ensure the system architecture is modular, scalable, maintainable, and suitable for future commercial expansion. Existing materials such as CAD files, STEP files, PCB layouts, schematics, firmware, architecture documents, and prior research outputs may be reviewed and referenced wherever technically beneficial, but the wearable hardware architecture, PCB systems, embedded firmware structure, BLE communication framework, mechanical integration, intelligent feedback workflows, and mobile application architecture will be independently redesigned and validated during this phase. The project includes the development of a wearable sensing platform capable of capturing and transmitting biomechanical movement data in real time through BLE communication while supporting session management, movement tracking, performance analysis, and export-ready data handling. The system will also support intelligent expansion features including haptic feedback integration, vibration actuator systems, movement comparison logic, adaptive correction workflows, coach-defined movement standards, AI-supported recommendation architecture, satisfactory performance feedback states, and future analytics capability. Mechanical development will focus on ergonomic wearable integration, stable sensor positioning, modular electronics mounting, actuator placement optimization, and user comfort during physical activity. The mobile application will be developed in Flutter for Android and iOS compatibility and will support wearable communication, session tracking, real-time monitoring, intelligent feedback visualization, coach workflows, export functionality, and future AI-assisted coaching integration. The primary objective of the project is to deliver a functioning prototype foundation suitable for testing, validation, and future expansion into a scalable intelligent biomechanical coaching and correction platform.

Key Requirements

- Real-time biomechanical movement monitoring
- Wearable sensor integration and embedded hardware development
- Custom PCB design with BLE communication capability
- Reliable BLE pairing, reconnection, and data transmission
- Flutter-based mobile application for Android and iOS
- Real-time haptic feedback and vibration actuator support
- Movement comparison and correction logic
- Coach-defined movement standards and AI-ready architecture
- Session tracking, performance summaries, and export functionality
- Support for CSV, JSON, and raw data export
- Modular, scalable, and maintainable system architecture
- Future AI, analytics, sensor, and haptic expansion capability
- Ergonomic wearable mechanical integration and stable sensor placement

The final delivery will include a functional prototype exosuit system with integrated sensors, PCB, firmware, and mobile app for movement tracking and session analysis. The primary objective of the project is to deliver a functioning prototype foundation suitable for testing, validation, and future expansion into a scalable intelligent biomechanical coaching and correction platform.

Chapter : 01

Research and Development

1.1 Approach

The Research and Development (R&D) phase of the Vantare VantageSuit™ project focuses on redeveloping and optimizing the wearable biomechanical platform for intelligent movement monitoring, posture analysis, corrective feedback, and future AI-assisted coaching workflows. While previous engineering materials may be reviewed and referenced where useful, the current development effort will proceed with a fresh system redevelopment strategy to ensure modularity, scalability, maintainability, and future expansion capability.

The R&D process begins with detailed investigation of wearable biomechanics, sensor fusion methodologies, posture correction systems, embedded sensing technologies, BLE communication structures, intelligent movement comparison logic, and real-time haptic feedback systems. The objective is to establish a wearable platform capable of monitoring body movement, identifying deviations from expected movement standards, and generating corrective feedback through vibration-based guidance mechanisms.

Existing project assets including CAD files, STEP files, schematics, PCB layouts, firmware structures, and technical documentation may be reviewed wherever technically beneficial. However, the hardware architecture, intelligent feedback systems, wearable integration strategy, movement analysis logic, and application communication framework will be independently redeveloped and validated during this project phase.

The R&D process includes:

- Biomechanical movement analysis research
- Sensor placement and wearable ergonomics analysis
- IMU, flex sensor, and force sensing evaluation
- Embedded controller and BLE architecture planning
- Real-time movement comparison methodologies
- Coach-defined movement standard architecture
- AI-supported recommendation structure planning
- Real-time haptic feedback logic
- Vibration actuator system analysis
- Adaptive feedback decision architecture
- Satisfactory performance state logic

Phases of R&D Section

Phase 1: System Requirement Analysis & Architecture Redevelopment

- Analyze objectives and workflow expectations
- Review previous engineering assets where beneficial
- Redevelop complete system architecture from scratch
- Define wearable monitoring and correction workflow
- Establish intelligent feedback system requirements
- Define modular hardware and software boundaries
- Prepare updated System Requirement Specification (SRS)
- Define future AI and sensor expansion capability

Phase 2: Sensor & Embedded System Research

- Research IMU-based movement tracking systems
- Evaluate flex sensor and force sensing integration
- Analyze wearable embedded controller platforms
- Develop BLE communication architecture
- Plan movement data acquisition framework
- Define reconnection and communication reliability behavior
- Develop filtering and calibration methodology
- Prepare future sensor expansion support

Phase 3: Intelligent Feedback & Haptic System Research

- Research vibration motor and haptic actuator systems
- Analyze haptic driver integration architecture
- Study actuator positioning for wearable correction feedback
- Develop real-time correction response logic
- Analyze latency expectations for feedback generation
- Develop adaptive feedback control methodology
- Define satisfactory performance state logic
- Prepare future multi-zone haptic expansion support

Phase 4: Movement Comparison & AI Recommendation Research

- Research movement comparison algorithms
- Develop coach-defined movement standard workflows
- Study AI-supported recommendation methodologies
- Define movement deviation detection systems
- Develop adaptive feedback decision logic
- Prepare future learning and analytics architecture
- Research movement quality threshold systems

Phase 5: Scalability, Validation & Optimization

- Validate modular system architecture
- Perform scalability analysis
- Conduct manufacturability planning
- Develop risk mitigation strategy
- Validate intelligent feedback workflows
- Prepare deployment readiness documentation
- Finalize future expansion architecture

Component In-House Inventory

Sr. No.	Component Name	Purpose	Estimated Qty
1	IMU Sensor (MPU6050 / BNO055)	Motion tracking and posture monitoring	6
2	Flex Sensor	Joint angle measurement	4
3	Force Sensitive Resistor (FSR)	Pressure and force detection	4
4	Vibration Motor (Coin / ERM Type)	Haptic feedback for posture correction	8
5	Haptic Driver IC (DRV2605)	Vibration motor control	2
6	ESP32 Development Board	Main controller + BLE communication	2
7	STM32 Controller (Optional for Final Phase)	Advanced processing for final system	1
8	BLE Module (if separate required)	Wireless mobile app communication	1
9	Li-Po Rechargeable Battery	Portable power supply	2
10	Battery Management System (BMS)	Charging and protection	2
11	Voltage Regulator Module	Stable voltage supply	2
12	Charging Module (TP4056)	Battery charging control	2
13	PCB Fabrication	Custom hardware board development	2
14	Connectors and Headers	Sensor and power connections	Multiple
15	Wiring Harness	Internal wearable connections	Multiple
16	Conductive Textile / Mounting Straps	Sensor placement and wearable structure	1 Set
17	Elastic Wearable Base Suit	Exosuit mechanical platform	2
18	Protective Enclosure	Electronics housing	2
19	OLED Display (Optional)	Local status monitoring	1
20	SD Card Module (Optional)	Local backup data logging	1

NOTE:
The Components listed in the component inventory are indicative and provided for preliminary planning and architectural understanding only. Final selection will be carried out during the detailed R&D phase based on validated functional requirements, power consumption analysis, form-factor constraints, availability, and system-level performance considerations.

Component Cost Table

Sr. No.	Component Name	Qty	Estimated Unit Cost (USD)	Total Cost (USD)
1	IMU Sensors	6	18	108
2	Flex Sensors	4	15	60
3	FSR Sensors	4	12	48
4	Vibration Motors	8	6	48
5	Haptic Driver IC	2	12	24
6	ESP32 Board	2	15	30
7	STM32 Controller	1	20	20
8	BLE Module	1	10	10
9	Li-Po Battery	2	20	40
10	BMS Module	2	8	16
11	Voltage Regulators	2	6	12
12	TP4056 Charging Module	2	5	10
13	PCB Fabrication	2	40	80
14	Connectors + Headers	Multiple	20	20
15	Wiring Harness	Multiple	25	25
16	Conductive Textile / Straps	1 Set	60	60
17	Wearable Base Suit	2	75	150
18	Protective Enclosure	2	20	40
19	OLED Display	1	10	10
20	SD Card Module	1	8	8

Estimated Total Cost = $819 USD

Estimated Total Cost with Contingency + Testing + Spare Components = $950–$1100 USD

DEVELOPMENT COSTING TABLE

Item	Estimated Cost
PCB Fabrication (1- Basic)	$200
PCB Fabrication (3- Plus)	$600
PCB Fabrication (5 -Premium)	$1000

Milestones and Deliverables

Milestones

- Requirement Analysis & System Architecture
- Sensor & Embedded System Research
- Intelligent Feedback & Haptic Research
- Movement Comparison & AI Workflow Planning
- System Validation & Scalability Planning

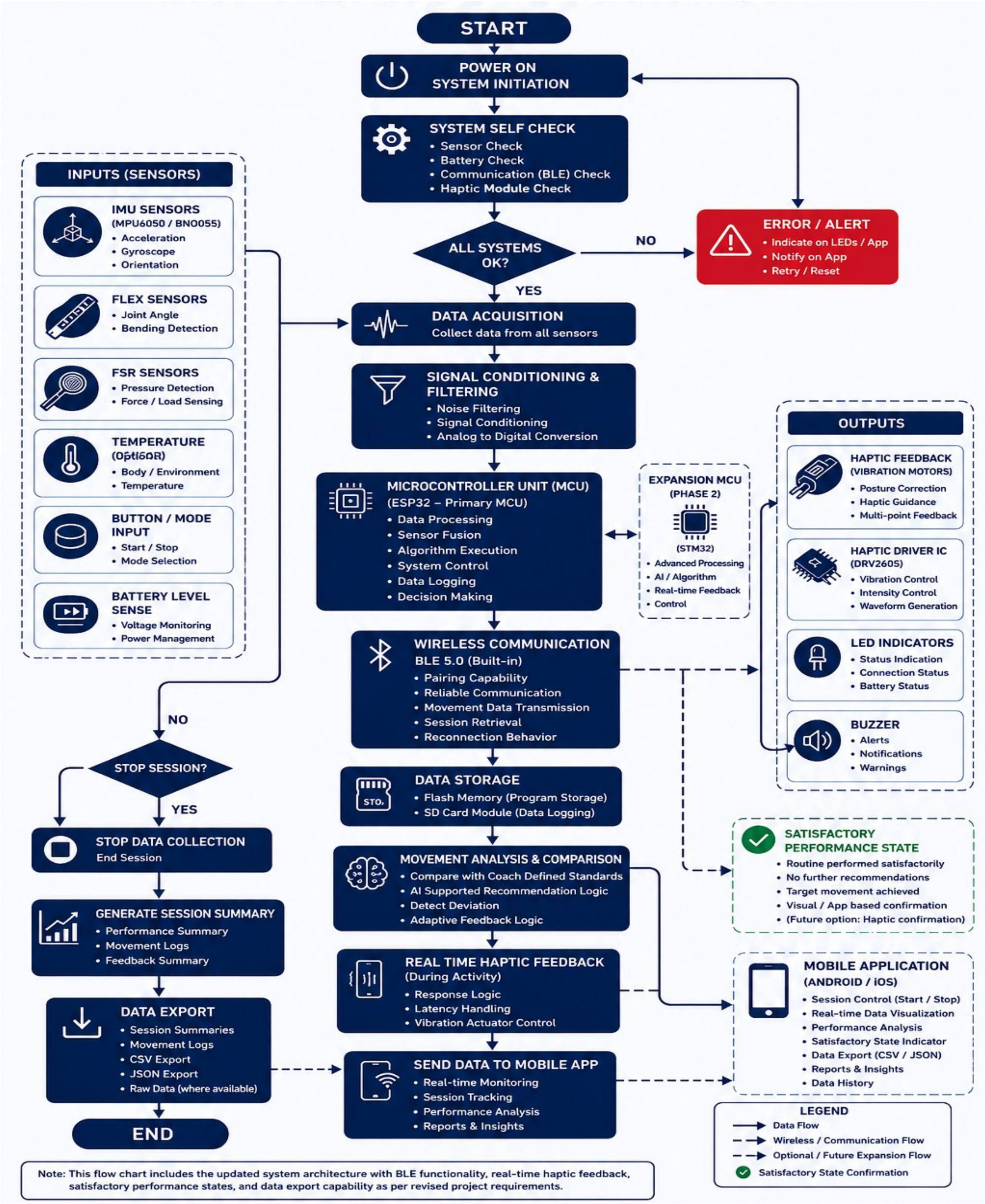
Deliverables

- System architecture documentation
- Sensor and embedded system research report
- BLE communication framework
- Haptic feedback and correction logic documentation
- AI-ready movement analysis workflow
- Scalability and validation report

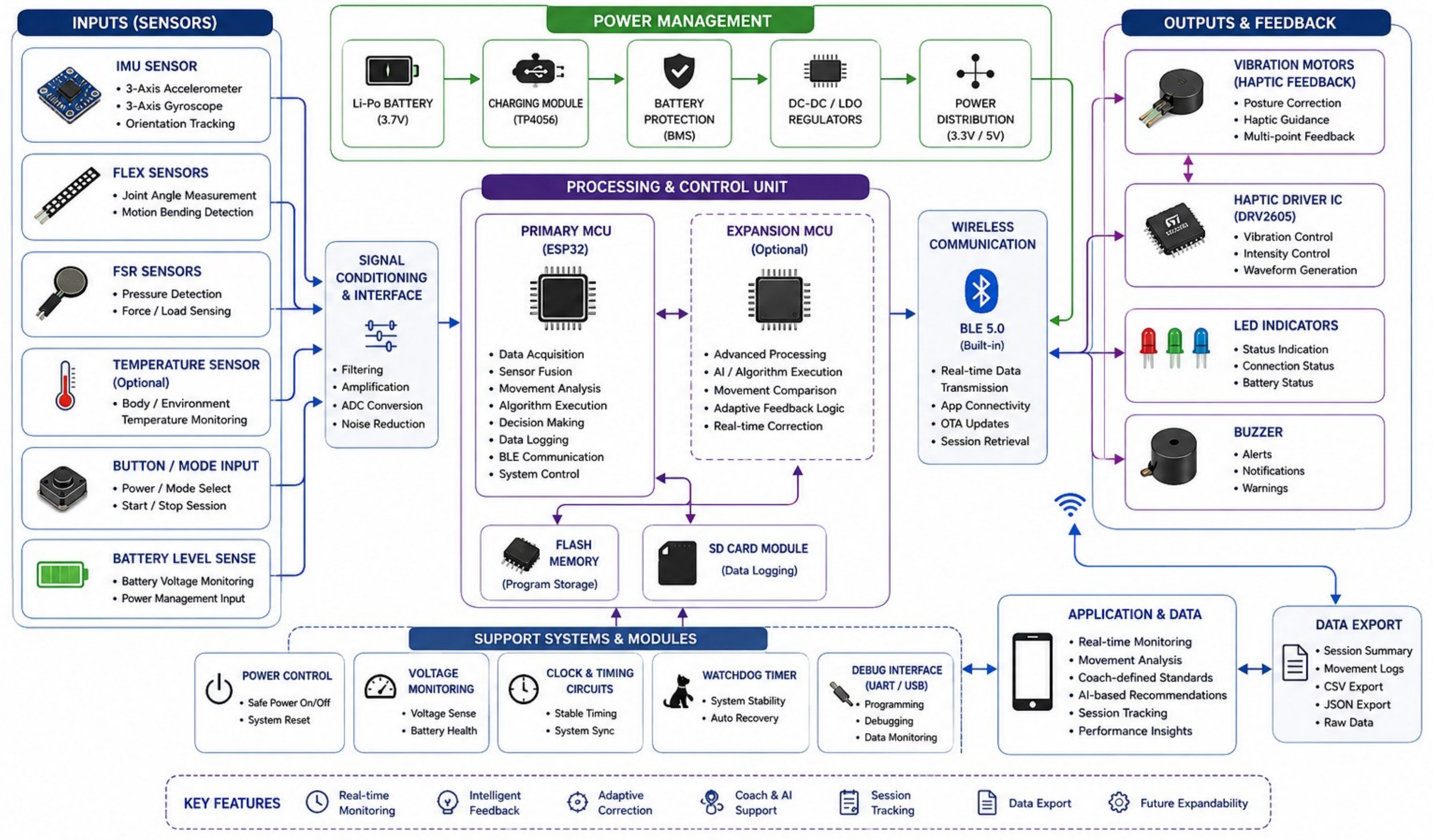
Outcomes

- Scalable wearable system architecture established
- Validated sensing and embedded framework
- Reliable BLE communication strategy prepared
- Intelligent corrective feedback workflow developed
- Future AI integration capability prepared
- Prototype-ready research foundation completed

Flow Chart


START
POWER ON
SYSTEM INITIATION
SYSTEM SELF CHECK
• Sensor Check
• Battery Check
• Communication (BLE) Check
• Haptic Module Check
ALL SYSTEMS OK?
NO
ERROR / ALERT
• Indicate on LEDs / App
• Notify on App
• Retry / Reset
YES
INPUTS (SENSORS)
IMU SENSORS
(MPU6050 / BNO055)
• Acceleration
• Gyroscope
• Orientation
FLEX SENSORS
• Joint Angle
• Bending Detection
FSR SENSORS
• Pressure Detection
• Force / Load Sensing
TEMPERATURE
• Body / Environment
• Temperature
BUTTON / MODE INPUT
• Start / Stop
• Mode Selection
BATTERY LEVEL SENSE
• Voltage Monitoring
• Power Management
DATA ACQUISITION
Collect data from all sensors
SIGNAL CONDITIONING & FILTERING
• Noise Filtering
• Signal Conditioning
• Analog to Digital Conversion
MICROCONTROLLER UNIT (MCU)
(ESP32 – Primary MCU)
• Data Processing
• Sensor Fusion
• Algorithm Execution
• System Control
• Data Logging
• Decision Making
EXPANSION MCU
(PHASE 2)
(STM32)
• Advanced Processing
• AI / Algorithm
• Real-time Feedback
• Control
OUTPUTS
HAPTIC FEEDBACK
(VIBRATION MOTORS)
• Posture Correction
• Haptic Guidance
• Multi-point Feedback
HAPTIC DRIVER IC
(DRV2605)
• Vibration Control
• Intensity Control
• Waveform Generation
LED INDICATORS
• Status Indication
• Connection Status
• Battery Status
BUZZER
• Alerts
• Notifications
• Warnings
WIRELESS COMMUNICATION
BLE 5.0 (Built-in)
• Pairing Capability
• Reliable Communication
• Movement Data Transmission
• Session Retrieval
• Reconnection Behavior
NO
STOP SESSION?
YES
STOP DATA COLLECTION
End Session
GENERATE SESSION SUMMARY
• Performance Summary
• Movement Logs
• Feedback Summary
DATA EXPORT
• Session Summaries
• Movement Logs
• CSV Export
• JSON Export
• Raw Data (where available)
END
DATA STORAGE
• Flash Memory (Program Storage)
• SD Card Module (Data Logging)
MOVEMENT ANALYSIS & COMPARISON
• Compare with Coach Defined Standards
• AI Supported Recommendation Logic
• Detect Deviation
• Adaptive Feedback Logic
REAL TIME HAPTIC FEEDBACK
(During Activity)
• Response Logic
• Latency Handling
• Vibration Actuator Control
SEND DATA TO MOBILE APP
• Real-time Monitoring
• Session Tracking
• Performance Analysis
• Reports & Insights
SATISFACTORY PERFORMANCE STATE
• Routine performed satisfactorily
• No further recommendations
• Target movement achieved
• Visual / App based confirmation
• (Future option: Haptic confirmation)
MOBILE APPLICATION
(ANDROID / iOS)
• Session Control (Start / Stop)
• Real-time Data Visualization
• Performance Analysis
• Satisfactory State Indicator
• Data Export (CSV / JSON)
• Reports & Insights
• Data History
LEGEND
Data Flow
Wireless / Communication Flow
Optional / Future Expansion Flow
Satisfactory State Confirmation
Note: This flow chart includes the updated system architecture with BLE functionality, real-time haptic feedback, satisfactory performance states, and data export capability as per revised project requirements.

Block Diagram



Chapter : 02

PCB Design

2.1 Approach

The PCB development section focuses on redesigning and optimizing a scalable embedded hardware platform capable of supporting wearable sensing, BLE communication, intelligent movement analysis, and real-time haptic feedback systems. While previous PCB files and schematics may be reviewed during development, the hardware architecture for this phase will be independently redeveloped from scratch to ensure long-term reliability, modularity, maintainability, and future intelligent system scalability.

The PCB system will support:

- Real-time movement sensing
- BLE-based communication
- Sensor data acquisition and processing
- Real-time haptic feedback control
- Vibration actuator integration
- Movement comparison support architecture
- Adaptive correction workflows
- Future AI-assisted expansion capability
- Future sensor and actuator scalability

The hardware architecture will prioritize:

- Modularity
- Scalability
- Maintainability
- Low-noise signal integrity
- Wearable compactness
- Battery safety and thermal protection
- Expansion accessibility
- Future intelligent correction support

Architecture definitions, communication structures, actuator pathways, sensor routing, and future expansion interfaces shall be fully documented during development.

Schematics

Phase 1 – System Architecture & Hardware Mapping

- Develop full wearable system architecture
- Define sensor-controller-feedback relationships
- Create modular expansion architecture
- Define BLE communication pathways
- Define actuator routing architecture
- Develop future AI expansion interfaces
- Document architecture definitions

Phase 2 – Sensor & Embedded Interface Design

- IMU sensor interfacing
- Flex sensor integration
- Force sensing interface circuits
- Embedded controller integration
- Filtering and signal conditioning design
- Sensor expansion support planning
- Data acquisition optimization

Phase 3 – Power & Protection Design

- Li-Po battery integration
- Charging and regulation architecture
- Reverse polarity and overcurrent protection
- Thermal safety design
- Wearable power optimization
- Expansion-ready power routing

Phase 4 – BLE Communication Hardware

BLE functionality shall support:

- Pairing capability
- Reliable communication behavior
- Movement data transmission

- Session retrieval capability
- Reconnection behavior
- Development activities include:
- BLE antenna optimization
- Wireless signal stability analysis
- Communication reliability validation
- Real-time data transmission testing

Phase 5 – Haptic Feedback Driver Architecture

- Vibration motor driver circuits
- DRV2605 integration
- Multi-channel actuator routing
- PWM-based intensity control
- Real-time feedback triggering support
- Latency-aware response pathways
- Safety-controlled actuator outputs
- Future adaptive feedback expansion capability

PCB Design

After schematic validation, the PCB Design section converts the logical circuit into a physical manufacturable board layout optimized for wearable performance, compactness, and future scalability. Since this exosuit is a wearable system, PCB design must prioritize size reduction, lightweight routing, flexible placement, low EMI, power efficiency, and user comfort.

Special attention is given to sensor placement strategy, routing isolation for analog and digital signals, battery safety, actuator current paths, and modular connector accessibility to support maintenance and field testing.

Phase 1 – Component Placement Strategy

This phase ensures logical and efficient positioning of all major hardware components on the PCB.

- Sensor placement optimization

- Controller placement planning

- Battery and charging module positioning
- Connector accessibility planning
- Wearable compact form factor study
- Future expansion space reservation

Phase 2 – PCB Routing and Signal Integrity

This phase focuses on creating stable electrical pathways while minimizing noise and interference.

- Analog and digital signal separation
- Ground plane optimization
- Power trace sizing
- High-current routing for actuators
- Noise reduction strategy
- EMI/EMC improvement planning
- Signal integrity validation

Phase 3 – DRC, ERC and Prototype Fabrication Readiness

This final stage ensures the PCB is fully ready for manufacturing and testing.

- Electrical Rule Check (ERC)
- Design Rule Check (DRC)
- Gerber file preparation
- BOM finalization
- Prototype fabrication readiness
- Assembly planning
- Testing preparation for hardware validation

Milestones and Deliverables

Milestones

- Complete schematic architecture finalized
- Sensor interface circuits validated
- Power management system approved
- BLE communication hardware finalized
- Haptic feedback driver architecture reserved
- PCB layout completed for prototype phase
- Future-ready expansion design completed
- ERC and DRC successfully passed
- Gerber files and BOM finalized
- Prototype fabrication package completed

Deliverables

- Complete schematic design files
- Full PCB layout design
- Sensor and controller interface circuits
- Power management and battery protection design
- BLE communication hardware design
- Haptic feedback expansion architecture
- Design Rule Check and Electrical Rule Check reports
- Gerber files for fabrication
- Final Bill of Materials (BOM)
- Prototype manufacturing package

Outcomes

- Functional prototype PCB ready for fabrication
- Stable hardware platform for Phase 1 deployment
- Future-ready architecture for Phase 2 intelligent correction system
- Reliable embedded platform for mobile app integration and real-time feedback system

Chapter : 03

App Development

3.1 Approach

The Vantare VantageSuit™ mobile application will act as the primary software interface between the wearable hardware platform, athletes, coaches, and future intelligent analytics systems. While previous application concepts may be reviewed during development, the application architecture for this phase will be redeveloped independently to ensure scalability, maintainability, modularity, and future AI-assisted coaching capability.

The application will be developed in Flutter to support Android and iOS platforms through a unified codebase. The app will support wearable communication, real-time movement monitoring, intelligent corrective feedback workflows, coach-defined movement standards, performance tracking, export-ready reporting, and future analytics expansion capability.

The application architecture shall support:

- BLE wearable connectivity
- Pairing and reconnection behavior
- Real-time movement monitoring
- Session tracking and retrieval
- Real-time haptic feedback support
- Coach-defined movement workflows
- AI-supported recommendation architecture
- Movement comparison systems
- Adaptive feedback logic
- Satisfactory performance feedback states
- Export-ready data handling
- Future learning and analytics integration

App Development Phases

Phase 1: Requirement Analysis & Architecture Redevelopment

- Define wearable communication workflow
- Define athlete and coach interaction structure
- Develop scalable app architecture
- Prepare BLE communication structure
- Define export-capable data architecture
- Define satisfactory performance state workflow

Phase 2: UI/UX Design

- Device pairing screens
- Session monitoring dashboard
- Live movement visualization
- Feedback status interface
- Positive completion state indicators
- Coach movement configuration screens
- Performance summary screens

Phase 3: Flutter Foundation Development

- Flutter project setup
- Navigation and state management
- Reusable component architecture
- Local storage setup
- Mock testing framework
- Scalable analytics-ready structure

Phase 4: BLE Communication Layer

BLE functionality shall include:

- Pairing capability
- Reliable communication handling
- Movement data transmission
- Session retrieval capability
- Reconnection behavior
- Development includes:
- BLE scanning and pairing
- Connection monitoring
- Reconnection handling
- Sensor packet receiving
- Hardware command support
- Communication reliability validation

Phase 5: Session Tracking & Real-Time Feedback

- Start/stop session workflow
- Live movement monitoring
- Real-time haptic feedback triggers
- Session logging

- Device status monitoring
- Adaptive correction logic integration
- Feedback latency handling

Phase 6: Performance Summary & Export System

System shall support export capability including:

- Session summaries
- Movement logs
- CSV export
- JSON export
- Raw data export where available
- Additional features include:
- Performance summaries
- Movement quality indicators
- Session history
- Progress visualization
- Export-ready reporting system

Phase 7: Intelligent Coaching & Recommendation Workflow

- Coach-defined movement standards
- AI-supported recommendation preparation
- Movement comparison workflows
- Adaptive correction logic
- Satisfactory performance feedback states
- Positive completion confirmation logic
- Examples may include:
- "Routine performed satisfactorily"
- "No further recommendations at this time"
- "Target movement achieved"
- Visual confirmation indicators

Phase 8: Testing, Optimization & Handover

- Android and iOS testing
- BLE reliability validation
- Session data verification
- Real-time feedback testing
- UI responsiveness optimization
- Bug fixing and cleanup
- Final documentation and handover

Milestones and Deliverables:

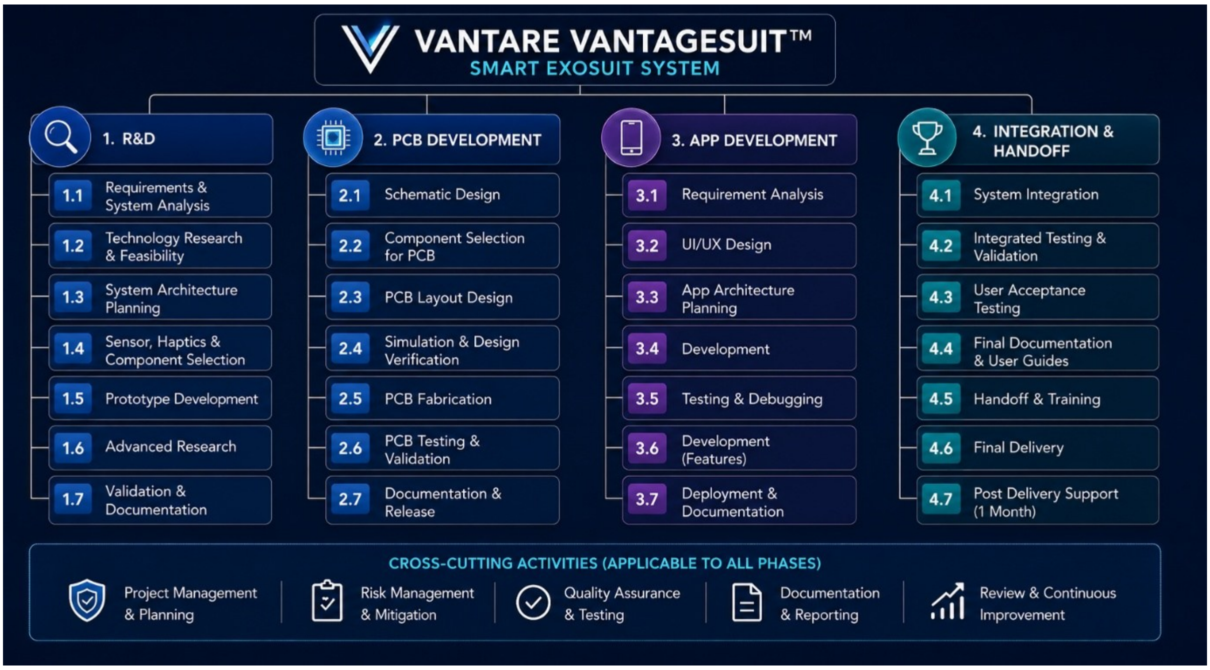
Milestone	Deliverables
Milestone 1: App Scope & Architecture	Final app flow, feature list, user roles, BLE communication plan, data structure
Milestone 2: UI/UX Design	Complete app screen designs for onboarding, device pairing, dashboard, live session, history, and reports
Milestone 3: Flutter Base App	Flutter project setup, navigation, reusable components, local storage, profile structure
Milestone 4: BLE/Hardware Integration Layer	Device scan, pairing, connection status, data receiving, command sending structure
Milestone 5: Session Tracking Module	Start/stop session, live data screen, session timer, battery/status display, local session logs
Milestone 6: Performance Summary Module	Session history, charts, basic movement summary, progress indicators
Milestone 7: Coach Module	Coach dashboard, athlete list, session assignment, basic training rule setup
Milestone 8: Testing & Handover	Tested Android/iOS app build, bug fixes, documentation, source code handover

Timeline

Note:

- It is important to note that the component procurement, sourcing, and logistics processes are not included within the defined R&D timeline. The provided schedule strictly covers engineering activities related to system analysis, debugging, optimization, and validation. Any delays or dependencies arising from component availability, supplier lead times, or procurement procedures fall outside the scope of this timeline.
- PCB fabrication will be carried out through JLCPCB, ensuring high-quality manufacturing and adherence to design specifications. The estimated lead time for fabrication and delivery is approximately 2-3 weeks.

Work Breakdown Structure



KPI (Key Performance Indicator)

A measurable value that indicates how effectively the project is meeting its objectives.

Rate: 95%

Disclaimer: The KPI rate reflects the targeted performance level under planned project scope and resources. Actual outcomes may vary depending on unforeseen technical or operational challenges.

Creep (Scope Creep)

Uncontrolled or gradual expansion of project scope beyond its original objectives.

Rate: 5%

Disclaimer: The creep rate represents an estimated buffer for minor deviations. Any expansion beyond this rate will require formal change requests and client approval.

Timeline and Budget Justification

Timeline Justification

The 10-week timeline is structured to reduce technical risk and keep each stage approval-based before moving forward. Week 1 is assigned to architecture approval so the system design, sensor placement, BOM, and integration plan are finalized before development begins.

Weeks 1–5.5 are allocated for R&D, hardware design, PCB schematics, layout, Gerber preparation, and fabrication package completion, as this is the most technical phase and requires validation before assembly.

Weeks 5.5–7.5 focus on mechanical prototype assembly, including suit integration, sensor mounting, haptic pod placement, battery placement, wiring, and fit testing.

Weeks 7.5–9 are reserved for hardware and app integration, where BLE communication, app pairing, movement tracking, session retrieval, and full system demo are tested together.

Weeks 9–10 are kept for final prototype validation, documentation, source-code handoff, ownership transfer, credentials transfer, and delivery of all purchased hardware to the client. Each milestone requires client approval to ensure controlled progress and avoid rework.

Budget Justification

The proposed project is justified by the comprehensive scope and technical depth of Design and Development of Vantare VantageSuit™. This rate covers the complete project lifecycle, including advanced research and system architecture definition, Component selection and validation, schematic and PCB design, wireless integration, App development, system testing, and preparation of detailed technical documentation. Sufficient time is allocated for iterations, debugging, and validation to ensure electrical safety, signal integrity, wireless reliability, and long-term performance in real-world. The budget also includes continuous client communication and post-delivery technical support for clarifications and minor refinements. The fixed-rate model ensures budget predictability, with advance payment supporting early resource allocation and final payment upon successful delivery, resulting in a balanced and value-driven budgeting framework.

Portfolio

Private & Confidential

Ultra-Slim Smart Healthband

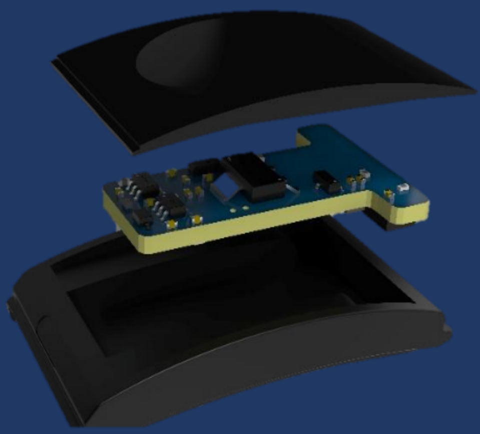
Muhammad Bilal Javid

Problem

Modernhealth monitoring systems are often bulky, power-consuming, and lack real-time connectivity. Users need a wearable device that provides accurate biometric monitoring, long battery life, and seamless smartphone/cloud synchronization while remaining comfortable for all-day use. Existing wearables also lack reliable OTA updates and touchless control features.

Goal

- Designan ultra-slim smart health band for accurate biometric monitoring.
- Enable gesture-based touchless control and haptic alerts.
- Ensure seamless BLE 5.0 connectivity using Nordic nRF52840 SoC.
- Support OTA firmware updates for long-term reliability.
- Provide ultra-low power performance (2+ days per charge).
- Maintain comfortable 5.5 mm slim design for all-day wear.
- Include precision sensors (heart rate, SpO_2, etc.) and optional display.



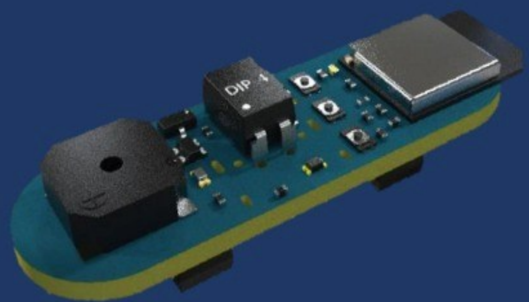
The Ultra-Slim Smart Health Band is a sleek, reliable wearable designed for modern health monitoring. It combines Nordic nRF52840 BLE 5.0, precision biometric sensors, gesture control, and haptic feedback for accurate real-time tracking. With ultra-low power optimization and OTA update support, it provides long-lasting usability. The device is ideal for remote healthcare, fitness, and medical IoT applications.

Smart Social WristBand

Muhammad Bilal Javid

Problem

Modernsocialinteractionsand networking events lack a discreet, low-latency method to broadcast social status, relying mainly on verbal communication and smartphone apps. Current solutions do not support secure proximity-based detection or privacy-preserving visibility. There is a need for a low-power wearable device that enables instant social cues with real-time synchronizationand minimal user effort.



Goal

- Develop a BLEwearable status band for easy social interaction.
- Display social/relationship status using color-coded LEDs.
- Enable real-time status sync via mobile app using Nordic nRF52840 SoC (BLE 5.0).
- Provide privacy mode and discreet vibration alerts.
- Support onboard button control and optional GPS mapping/chat features.
- Ensure User-Friendly, Non-intrusive, and wearable Device

My Status Band is a BLE wearable designed to make social interactions more engaging and approachable. By integrating the Nordic nRF52840 SoC, color-coded LEDs, vibration alerts, and real-time app synchronization, it offers a fun and non-intrusive way to signal social availability. With privacy protection and optional GPS-based features, the band supports safe and meaningful connections in events, meetups, and social communities.

Pilgrimage Campanion Device

Muhammad BilalJavid

Problem

Pilgrimsperforming Tawaf, Sa’i, and Tasbih often face challenges in tracking rituals accurately, accessing timely guidance, and managing orientation, especially in crowded or complex environments. Existing solutions may lack real-time responsiveness, GPS-based location accuracy, bilingual support, or reliable guidance, reducing convenience and precision.

Goal

- DevelopAL-Mueen, an STM32-based, event-driven pilgrimage companion.
- Integrate a SH1106 OLED display for clear visual guidance a DF Player audio engine for timely voice prompts
- Utilize a GNSS/GPS parser for precise location and time updates.
- Offer a bilingual experience with instant switching between English and Arabic for text and audio
- Ensure responsive interaction, smooth UI updates, and operational reliability in demanding environments
- Provide accurate guidance for Tawaf, Sa’i, and Tasbih using a cooperative state machine and interrupt-driven inputs

AL-Mueen delivers a responsive, bilingual, and GPS-enabled pilgrimage companion that guides users through Tawaf, Sa’i, and Tasbih with real-time visual and audio prompts. By combining STM32-based control, GPS positioning, an OLED display, and DFPlayer audio, the device ensures accurate, user-friendly, and consistent performance, enhancing the ritual experience for pilgrims.

Smart Insect Detection Camera System

Muhammad Bilal Javid

Problem

Accuratedetection and observation of insects is challenging due to their small size, rapid movement, and susceptibility to image distortion in conventional camera systems. Existing solutions often lack sufficient optical precision, stability, or real-time AI processing capabilities, limiting their effectiveness in research, agriculture, and biodiversity

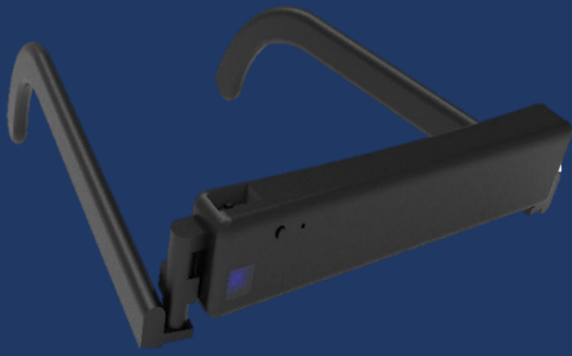
Goal

- Create a high-precision close-up imaging system for insects.
- Capture stable and distortion-free macro images.
- Integrate an embedded AI vision system for real-time detection.
- Use IMX219 8MP camera with a low-distortion lens setup
- Utilize MIPI-CSI interface for high-speed image transmission.
- Deploy RK3588 AI processor for on-device inference.
- Enable vibration-free imaging through optimized optical alignment.
- Provide AI/ML-ready real-time inference using TensorFlow Lite.
- Optimize system for field research and laboratory applications.

This project delivers a high-precision imaging and AI-powered insect detection system, combining optical accuracy with real-time machine learning. Using a low- distortion lens setup, IMX219 8MP camera, MIPI-CSI interface, and RK3588 processor, it enables stable close-up imaging and TensorFlow Lite inference. The system offers a reliable, scalable, and efficient solution for insect observation, research, and automated monitoring in both lab and field environments.

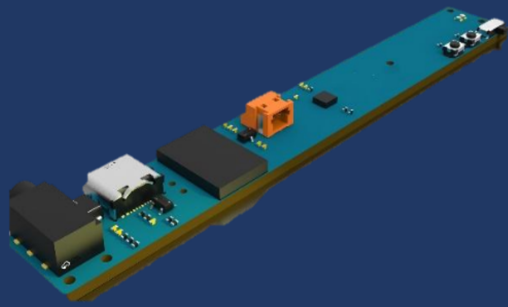
Smart Glasses

Muhammad Bilal Javid



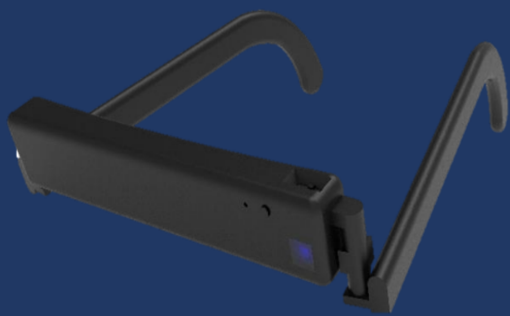
Problem

Individuals with physical disabilities often face challenges in interacting with computers or mobile devices due to limited mobility. Traditional input devices like mice and touchscreens are not always accessible, and existing assistive solutions may lack precision, responsiveness, wireless connectivity, or efficient powermanagement, limiting usability and convenience.



Goal

- Designand validate BLE-enabled smart glasses that control computer or mobile device cursors through head movements.
- Integrate a compact 4-layer PCB for a robust and space-efficient hardware platform.
- Utilize the MPU9250 gyro sensor for precise head movement tracking.
- Incorporate the BMD300 (nRF52832) Bluetooth module for reliable BLE connectivity.
- Implement a Battery Management System (BMS) for efficient charging, discharging, and thermal safety.



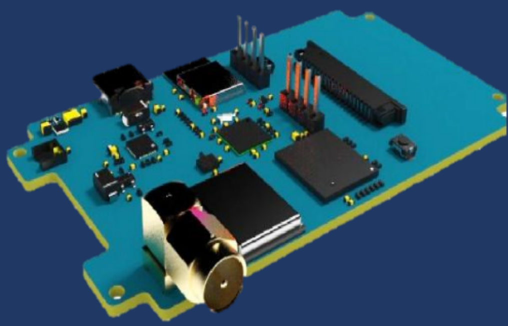
The BLE-enabled smart glasses provide an intuitive and accessible solution for handicapped individuals, enabling precise control of computers and mobile devices through head movements. By combining a compact 4-layer PCB, MPU9250 gyro sensor, BMD300 Bluetooth module, robust Battery Management System, and responsive interrupt-driven firmware, the device ensures seamless functionality, efficient power management, and user-friendly operation. This integration demonstrates a reliable and innovative approach to assistive technology.

Fitness Tracking System

Muhammad Bilal Javid

Problem

High-performanceracing enthusiasts require accurate and real-time lap and section timing for performance analysis. Existing solutions often lack seamless wireless connectivity, efficient on-device processing, or reliable data transfer to companion apps, limiting usability,precision, and post- session analysis.



G oal

- DevelopaBluetooth-enabled racing displays compatible with GPStrackerssuch as the Crossbox CBX30 and RaceBox Mini-S.
- Enable real-time calculation and recording of lap times, section times, and session statistics directly on the device.
- Ensure seamless Bluetooth connectivity for data transfer to the Crossbox mobile app.
- Incorporate the BMD300 (nRF52832) Bluetooth module for reliable BLE connectivity.
- Design a compact, low-power PCB optimized for signal integrity, EMI control, and efficient wireless communication.
- Implement an STM32-based control architecture for stable, real-time processing and reliable device performance.

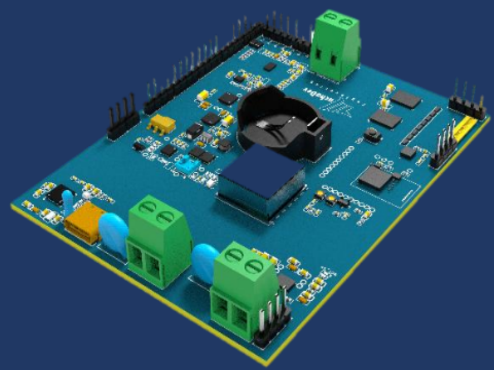
The Bluetooth-enabled racing display provides racing enthusiasts with a compact, reliable, and real-time timing solution. Its STM32-based architecture, optimized PCB design, and seamless Bluetooth connectivity allow accurate lap and section tracking, efficient on-device processing, and effortless integration with the Crossbox mobile app. This device demonstrates a robust and user-friendly approach to enhancing racing performance analysis.

Smart Shoe

Muhammad Bilal Javid

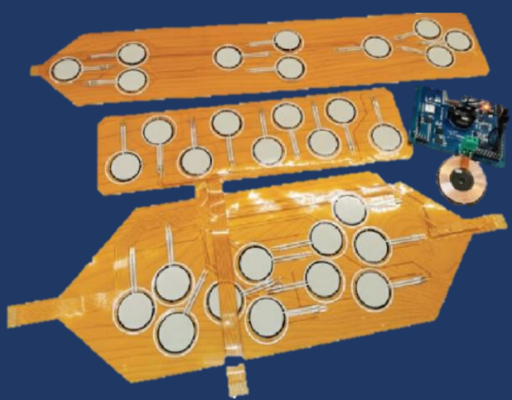
Problem

Patientswith mobility impairments, stroke recovery, musculoskeletal disorders, or diabetes often lack real-time monitoring of foot pressure and gait patterns. This can lead to abnormal pressure distribution, increased risk of falls, delayed rehabilitation, and diabetic foot ulcers. Traditional monitoring methods are often expensive, non-continuous, and not suitable for long-term daily use.



Goal

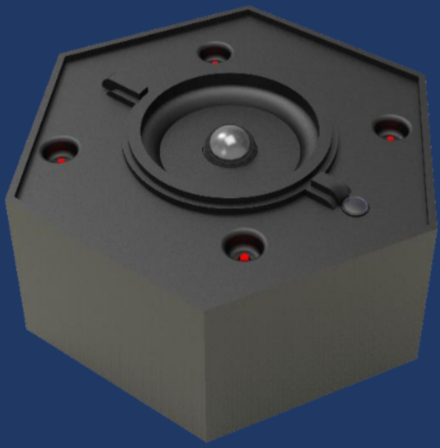
- Developasmart shoe with flexible FSR sensor PCB for accurate foot pressure detection.
- Enable real-time monitoring of pressure distribution and gait patterns.
- Use Nordic nRF52832 (BMD-300) BLE module for low-power wireless data transmission.
- Support long-term wearability and comfort with a compact shoe-integrated design.
- Include wireless charging, motion/magnetic sensors, and mobile dashboard support.



This smart shoe provides a medical wearable solution for real-time foot pressure monitoring and gait analysis using a flexible FSR sensor PCB and Nordic nRF52832 (BMD-300) BLE module. With motion/magnetic sensors, battery power management, and wireless charging, it ensures accurate detection, low-power wireless connectivity, and comfortable long-term wear. The system supports fall detection and diabetic foot monitoring, enabling preventive healthcare and rehabilitation through continuous tracking and data insights

Smart Fire Safety System

Muhammad BilalJavid

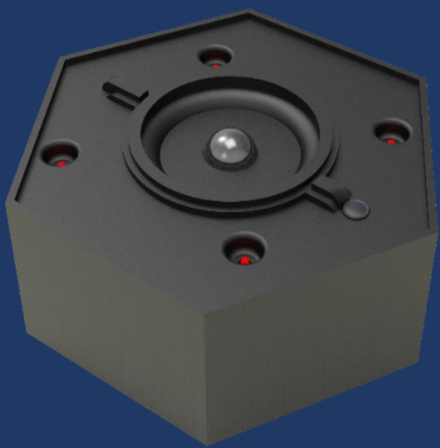


Problem

Firehazards andpoor indoor air quality pose serious risks to homes, offices, and industrial environments. Traditional fire safety systems often lack real-time monitoring, long-range communication, and intelligent alerts. Many solutions depend heavily on internet connectivity, have limited user interaction, and fail to provide early warnings for air quality degradation—leading to delayed responses andincreased damage or safety risks.

Goal

- Provide a reliable, wireless, and intelligent safety solution.
- Instantly detect smoke, fire, and temperature changes.
- Continuously monitor indoor air quality for early hazard detection.
- Operate reliably even without internet connectivity.
- Ensure long-range and stable communication using LoRa mesh networking.
- Offer an intuitive touchscreen interface for monitoring and manual control.
- Comply with EMI/EMC standards for safe and robust operation.



he Smart Fire Safety and Monitoring System successfully addresses modern safety challenges by combining advanced sensors, long-range wireless communication, and an interactive user interface. With real-time alerts, offline functionality, and robust design, it provides a dependable solution for homes, offices, smart buildings, and industrial environments. This system enhances early detection, improves response time, and significantly increases overall safety and reliability.

Smart Luggage

Muhammad Bilal Javid

Problem

Traditionalluggage lacksmodern security, real-time tracking, and weight awareness, making travel stressful and unpredictable. Zipper-based locks are vulnerable to tampering, lost luggage is difficult to recover, and travelers often face unexpected overweight baggage fees due to the absence of built-in weight monitoring and smartalerts.

Goal

- Design asmart, connected luggage system to enhance travel safety, convenience, and peace of mind.
- Integrate a magnetic fingerprint lock for secure, tamper-proof access.
- Integrate Arduino-based control for multiple peristaltic pumps.
- Incorporate an embedded digital weighing system to monitor luggage weight in real time.
- Provide remote monitoring and control through the Smartie mobile application.
- Deliver real-time alerts for location updates, arrival notifications, and overweight warnings.

The smart luggage solution successfully modernizes travel by combining secure biometric locking, real-time global tracking, and intelligent weight monitoring in a single connected product. With mobile app control and instant notifications, the system minimizes travel risks, prevents overweight baggage issues, and provides travelers with confidence and control throughout their journey. This project demonstrates a practical, scalable approach to intelligent travel gear.

Smart Humidifier

Muhammad Bilal Javid

Problem

Traditional aromadiffusers and incense burners often lack precise temperature control and advanced safety mechanisms, leading to risks such as overheating, inconsistent burning, and potential fire hazards. Many devices also operate without real-time monitoring or smart control, limiting user awareness and integration with modern smart home systems. Additionally, inefficient hardware designs result in higher power consumption and reduced

Goal

- Develop anIoT-enabled home automation system using ESP32.
- Enable remote control of lights, fans, and appliances via Wi-Fi or Bluetooth.
- Provide real-time monitoring and intelligent scheduling.
- Integrate a TFT LCD touchscreen for intuitive local control.
- Support offline operation for uninterrupted functionality.
- Improve energy efficiency through motion-based and single-light control.
- Enhance home safety with air quality monitoring and smoke alerts.
- Enable multi-zone lighting and fan speed control.
- Optimize system for field research and laboratory applications.

The ESP32-based IoT aroma diffuser successfully combines safety, precision, and smart functionality in a compact design. Through accurate temperature regulation, real-time monitoring, and intelligent protection mechanisms, the system ensures reliable and hazard-free operation. Its scalable architecture, low power consumption, and user-friendly interface make it well-suited for home, hospitality, and smart environment applications, offering a modern and safe solution for controlled aroma diffusion.

App of Wearable Device

Muhammad Bilal Javid

App Features

- Developed acompanionmobile app for a smart wristband prototype, built for investor demonstration and feature validation. Enabled seamless Bluetooth communication with
- the
  wristband.
- Displayed real-time sensor data and device status.
- Build a cross-platform mobile application for real-time monitoring and alert notifications. Design a wearable,
- ergonomic enclosure for user comfort and daily use.

App of Smart Home Controller

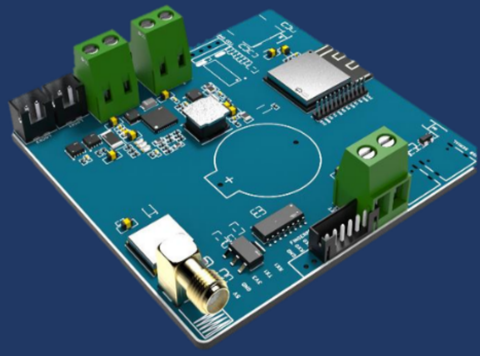
Muhammad Bilal Javid

App Features

- Developed amobileapp tointerface with a next-gen humidifier control system powered by ESP32
- Enabled real-time monitoring and control of humidifier settings via Wi-Fi and Bluetooth
- Supported manual and scheduled relay control for connected components
- Integrated sensor data visualization for humidity, temperature, and system status
- Designed for BMS-friendly operation **with** Modbus and VFC-based triggers

App of Smart Luggage

Muhammad Bilal Javid



App Features

- Acompanion mobileapp forasmart luggage system focused on security and convenience.
- Enabled fingerprint-based and app-based unlocking via Bluetooth.
- Supported remote device management and instant notifications.
- Included user authentication, role-based access, and automated report generation.
- Designed for scalability, reliability, and compliance monitoring.

App of Smart Device Admin Panel

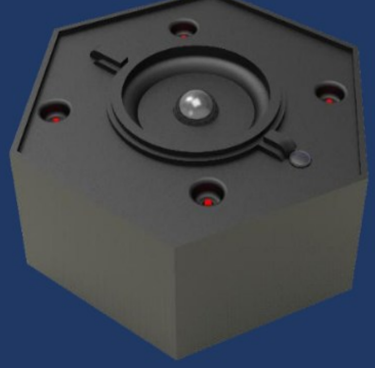
Muhammad Bilal Javid

App Features

- Designedan admin-facingweb app to manage and monitor smart home devices remotely.
- Enabled device registration, user management, and system diagnostics.
- Integrated with cloud backend for real-time status updates and troubleshooting.
- Provided usage analytics, firmware control, and alert notifications.
- Built with a responsive design for desktop and tablet use.
- Focused on reliability, control, and seamless backend

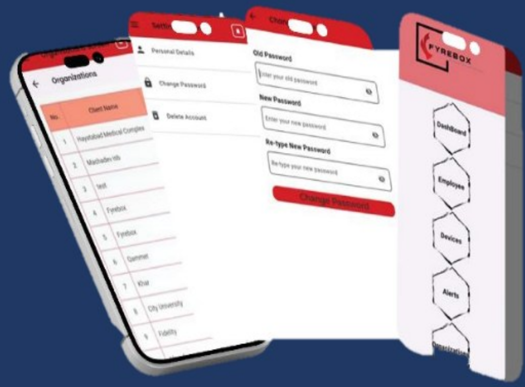
App of Fire Management

Muhammad Bilal Javid



App Features

- Developedacentralizedweb application for managing system Deployments monitoring.
- Implemented real-time dashboards for alerts, sensor data, and device status
- Supported remote device management and instant notifications.
- Included user authentication, role-based access, and automated report generation
- Designed for scalability, reliability, and compliance monitoring



App of Smart Humidifier

Muhammad Bilal Javid

App Features

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- Integrated sensor data visualization for humidity, temperature, and system status
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Client Feedback

Technology Officer for Electronic Product Development

Great to work with Muhammad and his team

★★★★★ Project Budget: $4,550

Pcb designer

Very professional team.

★★★★★ Project Budget: $5,250

PCB Layout

Muhammad and his team executed the project flawlessly with clear communication, quick responses, and efficient progress tracking. They handled changes smoothly and consistently aimed for fast solutions. Highly recommended for future projects.

★★★★★ Project Budget: $12,000

CAD design- Prototype

Muhammad and his team managed the project flawlessly from start to finish, with clear communication, fast responses, and excellent transparency via Google Docs. They efficiently handled changes and consistently delivered quick solutions. The client highly recommends them and would gladly work with them again.

★★★★★ Project Budget:$3,500

PCB and Electrical design for a photoreactor

I enjoy working with this group. They are talented.

★★★★★ Project Budget: $15,324

Electrical Engineer - IoT Kapton Heater Controller

Bilal and his team did good work. They are quick and communicate well, and are accommodating for any issues.

★★★★★ Project Budget: $3,658

ESP32 Developer Needed for Feature Enhancement

Excellent team. They are a decent sized team with lot of PCB development experience and firmware development. I worked with a team before and this team has exceeded my expectations. Will use again when I need. Thanks for all the help and assistance.

★★★★★ Project Budget: $$8,759

Arduino/RC32 Developer for GPS Based Alert System for Airsoft Game

It was a pleasure working with this team. Great service and product!

★★★★★ Project Budget: $5,823

Advance Existing Prototype - for Ultra Small Smart Flowmeter

Muhammad and his team delivered high-quality work by downsizing the PCB and choosing excellent components. Although firmware delivery took longer than expected, their accommodating nature and ability to bring in expert help led to a successful and satisfying project outcome.

★★★★★ **Project Budget: $18,000**

PCB Layout

Muhammad and his team managed the project flawlessly from start to finish, with clear communication, quick responses, and transparent progress tracking. They handled changes efficiently and delivered fast solutions. Highly recommended for future collaborations.

★★★★★ **Project Budget: $5,000**

Smart Alarm Clock With E-ink Display

Fast response time, good planning from the start and good support.
It was a pleasure to work with the team, and I hope to rehire soon again.

★★★★★ **Project Budget: $3,000**

Help Build Supplement Vending Machine Prototype

Muhammad Bilal and his team are very diligent, creative and hard working. their ability to be able to problem solve and create solutions for work moving forward has enabled the success of our project now and for the future. I would recommend to anyone.

★★★★★ **Project Budget:$ 5,000**

Schematic Design for Arduino Based Project with Stepper Motors

Exceptional product and service

★★★★★ **Project Budget: $17,312**

Automated Electric Shaver Development for Back Neck/Hairline

The team is extremely responsive and thorough. Highly recommend

★★★★★ **Project Budget: $3,658**

Mechanical Engineer Needed for Steel Bullet Trap Design with Electronics Integration

Highly skilled and responsive, delivering quality work with great attention to detail.

★★★★★ **Project Budget: $14,310**

Thank You!

200+	100+	30+	98%
Clients.	Project.	Countries.	Response rate

Thank you for your time and consideration. We would be happy to discuss our proposal further, address any questions you may have, and provide additional insights.

We look forward to your feedback and the opportunity to collaborate on the next steps.

CEO

MUHAMMAD BILAL JAVAID